Week 4

Case Study 1

Data for Questions 1 to 5

Delivery Delay Parameter Tuning

A logistics team is tuning a simple delay model used to estimate whether a shipment route is running slower than expected. To make the first gradient descent step auditable, the team reduces the training exercise to one parameter w, where smaller loss means the parameter is closer to the best delay setting. For one training example, the loss as a function of parameter w is

$$L(w) = (w - 4)^2.$$

The team uses gradient descent:

$$w_{t+1} = w_t - \eta \frac{dL}{dw}.$$

At the current step, $w_t = 1$ and $\eta = 0.1$. The team wants to know not only the next numerical value of w, but also why the sign of the gradient determines the direction of movement.

Question 1: What is $\frac{dL}{dw}$ at $w = 1$?

A) 6

B) -3

C) 3

D) -6

Correct Answer: D

Explanation: $\frac{dL}{dw} = 2(w - 4)$. At $w = 1$, the gradient is -6.

Question 2: What is the next value of w?

A) 1.3

B) 1.6

C) 0.4

D) 2.0

Correct Answer: B

Explanation: $w_{t+1} = 1 - 0.1(-6) = 1.6$.

Question 3: Why does w increase in this update?

A) The learning rate is negative.

B) The loss is already zero.

C) The gradient at $w = 1$ is negative.

D) The update ignores the derivative.

Correct Answer: C

Explanation: Gradient descent subtracts the gradient. Subtracting a negative value increases w.

Question 4: Which value of w minimizes this loss?

A) 1

B) 2

C) 3

D) 4

Correct Answer: D

Explanation: $L(w) = (w - 4)^2$ is minimized when $w = 4$.

Question 5: What is the role of the learning rate η?

A) It controls the update step size.

B) It stores the final target value.

C) It removes the need for a gradient.

D) It makes every loss value equal.

Correct Answer: A

Explanation: The learning rate scales how far the parameter moves in the negative gradient direction.

Case Study 2

Data for Questions 6 to 10

Energy Forecast Training Schedule

A campus energy group trains a neural network to forecast short-term electricity demand for academic buildings. The full dataset is small enough for classroom discussion, but the team still wants to compare the update behavior of batch, stochastic, and mini-batch gradient descent. It has 12 training records, and in one experiment the mini-batch size is 3. One epoch means that all 12 records have been used once. The group records the gradient from each mini-batch so it can reason about the tradeoff between noisy single-record updates and more expensive full-batch updates.

Question 6: How many mini-batches are processed in one epoch?

A) 3

B) 4

C) 6

D) 12

Correct Answer: B

Explanation: With 12 records and mini-batches of size 3, the number of mini-batches is $12/3 = 4$.

Question 7: Which method uses all 12 records for each parameter update?

A) Stochastic gradient descent

B) Mini-batch gradient descent

C) Batch gradient descent

D) Momentum-only update

Correct Answer: C

Explanation: Batch gradient descent computes each update using the full training set.

Question 8: Which method uses one record per parameter update?

A) Batch gradient descent

B) Adam without gradients or single-example updates

C) Full-batch averaging

D) Stochastic gradient descent

Correct Answer: D

Explanation: Stochastic gradient descent updates parameters using one example at a time.

Question 9: If the mini-batch gradients for four batches are $2, 4, 6, 8$, what is their average?

A) 5

B) 4

C) 6

D) 8

Correct Answer: A

Explanation: The average is $(2 + 4 + 6 + 8)/4 = 5$.

Question 10: Why might mini-batch gradient descent be preferred in practice?

A) It never uses gradients.

B) It balances noisy updates and computational efficiency.

C) It requires the loss to be exactly zero before every mini-batch update.

D) It updates only after training is complete.

Correct Answer: B

Explanation: Mini-batches often provide a practical tradeoff between noisy stochastic updates and expensive full-batch updates.

Case Study 3

Data for Questions 11 to 15

Queue Prediction Parameter Update

A service desk predicts queue length so it can decide when to open additional support counters. During training, successive gradients often point in similar directions but vary in magnitude from one mini-batch to the next. To smooth the parameter movement, the team trains a model using momentum. The update is

$$v_t = \beta v_{t-1} + \eta g_t, \qquad w_t = w_{t-1} - v_t.$$

At the current step, $v_{t-1} = 0.5$, $\beta = 0.8$, $\eta = 0.1$, and the gradient is $g_t = 3$. The previous parameter is $w_{t-1} = 5$. Here, v_t stores a running update direction, so the current step depends on both the new gradient and part of the previous movement.

Question 11: What is the new velocity v_t?

A) 0.5

B) 0.6

C) 0.7

D) 0.8

Correct Answer: C

Explanation: $v_t = 0.8(0.5) + 0.1(3) = 0.4 + 0.3 = 0.7$.

Question 12: What is the updated parameter w_t?

A) 4.3

B) 5.7

C) 4.7

D) 5.3

Correct Answer: A

Explanation: $w_t = w_{t-1} - v_t = 5 - 0.7 = 4.3$.

Question 13: What does the momentum term βv_{t-1} represent?

A) It removes all history from the update and uses only the current gradient.

B) It sets the learning rate to zero for every future update.

C) It replaces the loss function.

D) It carries part of the previous update direction.

Correct Answer: D

Explanation: Momentum keeps a decayed contribution from the previous velocity.

Question 14: If $\beta = 0$, what would v_t be for the same gradient and learning rate?

A) 0.3

B) 0.1

C) 0.2

D) 0.4

Correct Answer: A

Explanation: With $\beta = 0$, $v_t = \eta g_t = 0.1(3) = 0.3$.

Question 15: Why can momentum help optimization?

A) It makes gradients unnecessary.

B) It always increases the loss.

C) It can smooth updates across steps.

D) It forces all parameters to be equal.

Correct Answer: C

Explanation: Momentum can reduce oscillations and carry useful update direction across iterations.

Case Study 4

Data for Questions 16 to 20

Demand Prediction Step-Size Analysis

A retail analytics group trains a demand prediction model whose parameters do not all receive gradients of the same scale. To understand adaptive learning rates, the team isolates one parameter and tracks how its accumulated squared gradient changes the update size. For this parameter, the current gradient is $g = 4$. In an AdaGrad-style update, the accumulated squared gradient is

$$r_t = r_{t-1} + g^2.$$

At the current step, $r_{t-1} = 9$. The learning rate is $\eta = 0.5$, and the simplified update magnitude is

$$\frac{\eta g}{\sqrt{r_t}}.$$

This simplified calculation helps the team compare a plain gradient step with an adaptive step that becomes smaller when the accumulated squared gradient is large.

Question 16: What is r_t?

A) 16

B) 25

C) 13

D) 9

Correct Answer: B

Explanation: $r_t = 9 + 4^2 = 25$.

Question 17: What is $\sqrt{r_t}$?

A) 3

B) 4

C) 5

D) 6

Correct Answer: C

Explanation: Since $r_t = 25$, $\sqrt{r_t} = 5$.

Question 18: What is the update magnitude?

A) 0.2

B) 0.4

C) 0.8

D) 0.6

Correct Answer: B

Explanation: $\eta g/\sqrt{r_t} = 0.5(4)/5 = 0.4$.

Question 19: What is the main idea behind AdaGrad-style scaling?

A) It ignores all previous gradient information while claiming to adapt steps.

B) It requires the same update for every parameter.

C) It removes the need for a loss function.

D) It adapts the step size using accumulated squared gradients.

Correct Answer: D

Explanation: AdaGrad scales updates using accumulated squared gradients, giving parameter-specific effective step sizes.

Question 20: Which optimizer combines momentum-like averaging with adaptive scaling?

A) Adam

B) Plain batch gradient descent

C) Stochastic gradient descent without memory

D) A zero-learning-rate update

Correct Answer: A

Explanation: Adam uses moving averages of gradients and squared gradients.